IILM UNIVERSITY

INDUSTRY INTERNSHIP

Titanic Survival Prediction

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Enrollment Number: 2410030084

Program & Semester: B.Tech CSE — Semester 5

Organization: CodSoft

Internship Overview

Organization

CodSoft
Virtual internship in Data Science

Project

Titanic Survival Prediction
Binary classification

Duration

1 August 2026 – 31 August 2026
4-week virtual internship

Objective: build a machine learning model to predict whether a Titanic passenger survived.

Work included data preprocessing, Exploratory Data Analysis (EDA), model training and evaluation.

The internship provided practical exposure to data science and machine learning workflows.

Organization Profile

CodSoft

Technology and software development organization offering internship and learning opportunities.

Provides virtual programs in areas such as Data Science, Machine Learning, Web Development, Python and Java.

Uses a project-based learning approach, with interns completing assigned tasks within a specified timeframe.

Aims to bridge the gap between theoretical knowledge and practical, industry-relevant experience.

Problem Statement & Objectives

Problem Statement

Analyze the Titanic passenger dataset to identify patterns associated with survival and build a binary classification model where Survived is the target variable (0 = deceased, 1 = survived).

Project Objectives

- Clean and preprocess data containing missing values.
- Perform EDA to study survival patterns by age, gender and class.
- Train a Logistic Regression model.
- Evaluate predictions on unseen/test data.

Project Scope & Dataset

Domain

Data Science & Machine Learning

Learning type: Supervised Learning

Task: Binary Classification

Key Features

Pclass — Ticket class
Sex — Gender
Age — Passenger age
Fare — Ticket fare
Cabin / Embarked — Additional passenger information

Target

Survived

0 → Deceased
1 → Survived

Goal: maximize prediction accuracy on unseen data.

Technologies & Tools Used

Python

Programming and implementation

Jupyter Notebook

Interactive coding and visualization environment

Pandas

Data loading, manipulation, missing-value handling and feature selection

NumPy

Numerical operations and array handling

Matplotlib & Seaborn

Charts, distributions and heatmaps for EDA

Scikit-Learn

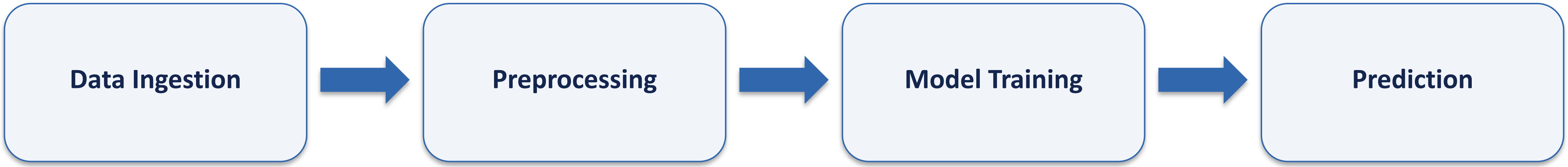
Logistic Regression and evaluation metrics

Methodology — CRISP-DM

1. **Business Understanding**
   Define survival prediction goal
2. **Data Understanding**
   Explore passenger attributes and correlations
3. **Data Preparation**
   Clean data and handle missing values/outliers
4. **Modeling**
   Apply Logistic Regression
5. **Evaluation**
   Measure accuracy on validation/test data

System Architecture

Data Science Pipeline



Raw Titanic data is transformed through preprocessing and model training to produce survival predictions.

Results & Key Findings

~80%

Test Accuracy

Logistic Regression
Binary Classification

EDA Findings

- Female passengers had a significantly higher survival rate.
- 1st Class passengers had a higher chance of survival than 3rd Class.
- Children under 10 showed a higher survival rate.

Model Evaluation

- Accuracy was approximately 80% on the test data.
- Confusion Matrix showed a balanced number of false positives and false negatives.
- Results indicate a stable predictive model.

Learning Outcomes & Conclusion

Gained practical experience in the end-to-end data science workflow.

Learned to preprocess real-world datasets and handle missing/inconsistent data.

Strengthened skills in Exploratory Data Analysis and data visualization.

Applied Logistic Regression for a practical binary classification problem.

Learned to evaluate a machine learning model using accuracy and a confusion matrix.

The internship helped connect academic concepts with hands-on project implementation.

THANK YOU

Titanic Survival Prediction

Sanyaritu Bhatia | 2410030084 | B.Tech CSE — Semester 3